

OMER KOC

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Summary

Data Scientist pursuing a Master's in Data Science with expertise in Python, SQL, R, and AWS. Skilled in machine learning, deep learning, and statistical analysis, focusing on predictive modeling, visualization, and scalable data solutions. Experienced in analyzing large datasets and delivering actionable insights to enhance decision-making.

Education

Northeastern University **Jan. 2023 – Jun. 2025 (Expected)**
M.Sc. in Data Science. GPA: 3.95 Boston, MA
Relevant Coursework: Algorithms, Unsupervised/Supervised Machine Learning, Database Management Systems

Istanbul Technical University **Sep. 2016 – Jun. 2021**
B.Sc. in Industrial Engineering. High Honor List. GPA: 3.72 Istanbul, Turkey

Selected Projects

IMDb Movie Review Sentiment Analysis | *Python, Deep Learning, NLP* 🐙 **May 2023**

- Built a CNN-LSTM hybrid model for NLP tasks with embeddings and dropout, achieving 90.18% accuracy.
- Preprocessed data with HTML removal, stopword elimination, and lemmatization for consistency.

Design & Implement a Relational Database: Bird Strike | *R, MySQL* 🐙 **May 2024**

- Developed a relational database in MySQL for bird strike incidents, incorporating advanced data manipulation with R.
- Developed advanced SQL queries for data extraction and transformation in MySQL

Intraday Electricity Market Price Forecast Using Machine Learning | *Python, Machine Learning* 🐙 **Jun. 2021**

- Achieved a \$1.5 MAE using a CNN-LSTM hybrid model, leveraging deep learning techniques to outperform Bi-LSTM and GRU models, demonstrating expertise in time-series analysis.
- Used Facebook's Prophet to preprocess over 50 features, enhancing model accuracy with advanced forecasting techniques.

Experience

MATREU **Sep. 2024 – Dec. 2024**
Data Analyst Austin, TX

- Automated data workflows and reporting, increasing efficiency and ensuring timely delivery of insights to the team.
- Analyzed over 50 product performance metrics, identifying opportunities to improve sales and optimize product listings, leading to a 10% increase in monthly revenue.

Khoury College of Computer Sciences **Dec. 2023 – Jun. 2024**
Graduate Teaching Assistant Boston, MA

- Held weekly office hours to provide support on SQL, Python, Data Modeling, and tools such as MongoDB and Spark.
- Evaluated student code for accuracy and adherence to requirements and assigned grades for projects and exams.

Dynflux **Jan. 2022 – Dec. 2023**
Data Science & Analytics Specialist Boston, MA

- Developed predictive models using Python and machine learning techniques (Random Forest, XGBoost) to forecast customer demand and optimize pricing strategies, leading to a 15% increase in sales efficiency.
- Performed data wrangling and feature engineering on large-scale datasets (SQL, Pandas) to improve model accuracy and streamline data pipelines for real-time analytics.

Turkcell **Dec. 2020 – Jan. 2022**
Data Analyst Istanbul, Turkey

- Streamlined data workflows with Python and SQL, boosting forecast accuracy by 25% and leveraging ML for insights.
- Automated data extraction processes using SQL and Python, reducing manual effort by 30% across reporting tasks.

Leadership / Extracurricular

New England Turkish Student Association (NETSA) **Sep. 2023 – Present**
Vice President

- Increased the organization's treasury by 45% compared to the previous year through strategic grant and sponsorships.
- Facilitated networking events, strengthening the connection between Turkish and American student communities.

Technical Skills

Languages & Tools: Python, R, SQL, C++, MongoDB, Amazon Web Services (AWS), Microsoft Azure, MS Office
Expertise: Data Analysis, Statistics, Algorithms, Machine Learning, Deep Learning, Optimization, Relational Databases